

Anonymous Online Support: Investigations of Identity and Heterogeneous Groups in Online Recovery Support

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ABSTRACT

How does anonymity impact perceptions of the effectiveness of support in online recovery communities? Prior work has shown how the multifaceted concept of identity plays an important role in support seeking, but its effects with respect to support providers are still unclear. I ground my investigation of interpretations of 12-Step anonymity in online recovery to demonstrate the effects of different facets of anonymity and 12-Step values on perceptions of supportive interactions using a controlled online experiment. I present a novel transposition of person-centered support and demonstrate its utility outside of dyadic interactions. Finally, I discuss implications for the design of mixed online health communities and implications for future emotional support and anonymity research.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; *HCI design and evaluation methods; Empirical studies in HCI*;

KEYWORDS

Anonymity, recovery, online health community, peer support, verbal person-centeredness

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1 BACKGROUND

Human-Computer Interaction (HCI) scholars have worked to improve health outcomes for people living with various chronic health conditions using a variety of computational methods [15, 64, 72]. These works target their investigations towards online health communities (OHCs) and explore how people maintain social support networks to cope with health-induced stressors [14, 17, 27, 33, 57, 61]. Within recovery from Substance Use Disorders (SUDs), positive engagement with one's online recovery network is important to sustained online recovery [9]. In this paper, I present findings from

my controlled online survey experiment investigating the impact of different facets of anonymity and 12-Step values on perceptions of supportive interactions. My contribution to this space is threefold:

- (1) I empirically investigate the outcomes of observed behaviors and norms identified in prior work regarding anonymity in online recovery communities.
- (2) I present the effects of cultural differences between organizations within an overarching stigmatized group, people in recovery, with respect to support outcomes.
- (3) I extend Verbal Person-Centeredness (VPC) theory by demonstrating its utility in analyzing the effects of online support outside dyadic interaction.

Pathways to long-term recovery are not all the same with numerous maintenance programs and mutual-help organizations, such as 12-step recovery groups like Alcoholics Anonymous (AA) or Narcotics Anonymous (NA), SMART Recovery, or cognitive behavior therapy [43]. [58]. Managing anonymity while building peer support networks is a salient theme across many investigations of 12-Step online recovery [30, 58, 73]. While 12-Step fellowship is strongly predictive of long-term recovery status [48], its emphasis on powerlessness, religious aspects, and demand for readiness for change are often cited as barriers for participation [45]. My work presents the first attempt to characterize the effect of 12-Step interpretations of anonymity between alternative recovery groups online.

Self-disclosure, or “revealing personal information” [60], is driven by the need for social connection and can provide a number of benefits [4, 8, 16]. Users who choose to self-disclose online must navigate the tensions associated with perceptions of risk [24, 53]. I build on prior findings describing how different actors enact anonymity in online 12-step to navigate these tensions [58] and evaluate the impact of anonymity violations in light of ambiguous norms and traditions in online recovery spaces.

A large body of this work defines anonymity as concealing one's identity [6, 20, 25, 47, 66]). Using this definition of anonymity, recent work explores how the desire for anonymity is context dependent [21] and subject to temporal changes [22, 58] similar to the aforementioned tensions associated with self-disclosure. Other work argues this dichotomous definition doesn't capture the nuances of anonymity and instead proposes a relational model centered around the context of an interaction, the relationship between actors, and the content of the disclosure [22]. I find direct connections between interpretations of anonymity in online 12-step and Eklund et al.'s micro facets (factual, social group, and physical) of anonymity [58]. Using these works, I ground the following four RQs in both theory and practice:

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RQ1.What effect does disclosing factual/physical identity have on perceptions of online recovery support?

RQ2.What effect does disclosing social in/out-group identities have on perceptions of online recovery support?

RQ3.What effect does joint disclosures of identity have on perceptions of online recovery support?

RQ4.What effect does 12-Step participation have on the impact identity has on support perceptions?

The "message perception paradigm" focuses on measures local to instances of support and has been praised for its ability to isolate and observe specific message features in supportive interaction [37]. Prior work builds on the assumption that some forms of supportive communication are qualitatively better than others [23] and finds that responses from support seekers vary with the quality of supporting messages [69]. I focus my investigation on perceived effectiveness (PE) in light of research showing it is causally antecedent to actual effectiveness (AE) [11, 18, 62]. I utilize a simplified version of Goldsmith et al.'s three-factor model as a measure of PE due to its notoriety, low cognitive burden on participants, and evaluation over multiple supporting dimensions (problem-solving, relational assurance, and emotional awareness) [29]. My approach is similar to methods utilized in prior work [28, 39], but is sensitive to evidence that these supportive dimensions signal distinct expressions of emotional support [29].

Other work has found verbal person-centeredness (VPC), defined as "an awareness of and adaptation to the affective, subjective, and relational aspects of communication contexts" [13], to be a valuable objective measure of emotional support quality [38]. Low (LPC), moderate (MPC), and high (HPC) major levels of VPC operationalize the degree to which supporting messages acknowledge and legitimize the feelings and perspectives of others [10, 11, 31, 32]. Most closely related to my work are investigations of factual, physical, and social group identity cue's impact on message production finding that identifiable profiles and in-group identities tended to receive higher VPC responses [23, 32, 46]. Research has not yet explored how environmental factors affecting message production may affect the underlying interpretations of person-centered messages and should be considered:

RQ5.What effect does VPC have on perceptions of online recovery support?

2 METHODS

Similar to prior investigations of identity markers in online support [23], I used a controlled online survey experiment to quantify the extent to which enacted anonymity, or conversely, the presence of identity cues affects the PE of supportive messages in online recovery. All research activity was reviewed and approved by our university IRB.

2.1 Participants and Recruitment

I used Prolific.co to recruit participants due to its superior data quality when compared to similar crowd-sourcing platforms [41, 54] and out-of-the-box support for several hundred demographic pre-screeners [3]. Specifically, I recruited participants who had self-reportedly been in individual therapy for alcohol use because they may be more familiar with expectations in online recovery support.

Data collection lasted from 2022-09-14 to 2022-09-27 and at the time 780 active Prolific users were eligible to participate out of 115,265.

Overall, 150 participants (60% male, 39% female, and 1% non-binary) completed the entire study and five started but did not complete it. I conducted a pilot with the first 25 participants to determine an estimated study duration of 12 minutes and set compensation at \$15/hr (\$3.12 per submission). All participants were from the United State with 78% being White/Caucasian, 10% Black/African American, and 13% representing six diverse ethnicities. The average age of participants was 40 (10.25) years. 33% of participants attended a 12-Step meeting in the last month; I used this distinction to characterize participants familiar with and in support of 12-Step norms.

2.2 Experimental Design

The survey was hosted on Qualtrics and distributed to participants via Prolific. Fig 1 illustrates the survey's flow comprised of a pre-screener question, consent form, recovery demographic questions, an example question, three main study questions, and general demographic questions. The main survey body was composed of three interrelated, sequential experiments relating to my fundamental research questions regarding factual/physical anonymity, social anonymity, and their co-occurrence. Participants were instructed to read each post from a mocked online recovery community and assist a prospective support provider by rating the PE of two drafted supportive comments.

2.3 Analytic Methods

Participants evaluated the perceived supportiveness of comments across three distinct but related dimensions (problem-solving, relational assurance, and emotional awareness) using three 100-point semantic differential slider scales (hurtful - helpful, unsupportive - supportive, insensitive - sensitive). In light of violations of normality, I analyzed PE ratings along each supporting dimension for non-example questions using Bayesian modeling techniques [49]. Namely, I leverage recent developments in Ordered Beta Regression using the R package `ordbetareg` to accurately model the bounded nature of slider data [44]. I modeled the dependence between my within-participant conditions using multilevel modeling techniques to investigate the interaction effects between factual/physical anonymity and social in/out-group conditions, and improve my models' ability to generalize to the greater population [52]. I include all experimental conditions, applicable participant demographic features, and interactions relevant to my RQs based on theoretical significance and experimental constraints [65]. I triangulate these results with a thematic data-driven analysis [51] on all free text responses from participants. I captured individual units of meaning with open codes and clustered them based on thematic affinity. Finally, I collaborated with colleagues from my lab to analyze qualitative themes until a set was established that effectively characterize participants' preferences in online recovery support.

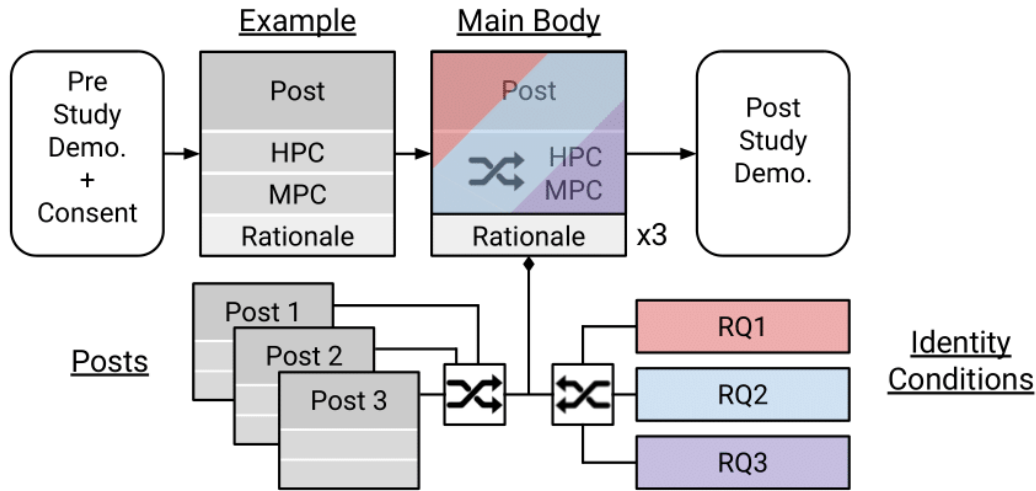


Figure 1: A figure describing the experimental layout of the online survey. This mixed-subject design investigates identity conditions within RQs as between-subjects factors and person-centeredness and identity conditions between RQs as within-subjects factors. Participants were randomly assigned to an identity condition within experiments where each experiment relates directly to RQ1-3. The post associated with each condition and the order of the conditions was randomized.

2.4 Identities, Posts, and Comments

I direct my investigation of anonymity at contextually relevant disclosures of identity cues to symbolize different anonymity violations. I outline three distinct identify facets and contextualize them in recovery norms and traditions.

- **Factual/Physical:** People in online 12-step recovery often associate unidentifiability with factual and physical anonymity and an analysis of this "sharing name" and "sharing face" behavior shows that the majority of users enact these kinds of anonymity online [58]; as such, I analyze both facets together. **Factually/physically identifiable comments contained a first name, last name, and photo that mimicked a linked social media profile.** Notably, one's name and face disclose inherent social group identities (i.e., race and gender) [22]. Therefore I randomized the presented gender and race in the linked profile ¹.
- **Social In-Group:** Some people in online 12-step recovery associate anonymity with not disclosing recovery status [58], but this desire is sometimes at odds with goals for support provisioning [59]. **Socially identifiable in-group comments explicitly disclose the helper's identity as a person in recovery.**
- **Social Out-Group:** Similarly, out-group identities are often used to measure the effects of their in-group counterparts [46, 55]. Within 12-step groups, social group anonymity is considered a means of equal rights for the diverse set of backgrounds represented in recovery groups [58]. Historic differences between clinicians and 12-step participants can accentuate negative attitudes between both groups [45], but prior work hasn't investigated how this relatable out-group identity affects supportive interaction. **Socially identifiable**

out-group comments explicitly disclose the helper's identity as a medical professional in the space of recovery.

With these three identity conditions in mind, I form the basis for my empirical investigation of anonymity in the context of online recovery through six experimental conditions: 1) fully anonymous, 2) factually/physically identifiable, 3) social in-group, 4) social out-group, 5) identifiable social in-group, and 6) identifiable social out-group.

I collaborated with a member of a 12-Step community to generate example posts based on our combined experiences in recovery and OHCs. To simulate a real OHC, the posts were visually designed to resemble the interface for Reddit's r/recovery with embedded sliders below each comment. I created an HPC and MPC comment for each post; the main difference is characterized by an explicit elaboration of the seeker's feelings compared to an attempt to smooth over those feelings by acknowledging the situation. Four trained research volunteers unfamiliar with the details of the present study reviewed these comments and provided independent VPC ratings to validate my assessment. I supported the randomization described in Fig. 1 by creating three versions of the same supporting comment: 1) no textual identity disclosures, 2) an in-group identity disclosure, and 3) an out-group identity disclosure.

3 RESULTS

I collected 900 PE ratings of helpfulness, supportiveness, and sensitivity for online supportive interactions across six identities and two person-centered conditions. 63% of participants provided 303 free text responses on a wide variety of topics (i.e., PC elements, 12-step, and personal recovery stories). I generated five salient major themes from 484 codes to provide a rich grounding from my quantitative results. Participants found interactions to be overwhelmingly helpful, supportive, and sensitive for all conditions with an average

¹Similar AI-generated faces were selected across eight gender/race pairs [2] using popular ethnically represented surnames [12]

rating of 78.5 (17.9). I sampled data using brms defaults to fit a multilevel ordbetareg model for each supportive dimension (see Table 1 for model coefficients)).

3.1 Factual/Physical and Joint Self-Disclosure

RQ1 directly relates to the first-order effects of factual/physical identifiability. As such, I interpret its coefficient in the main effects model while considering the interaction effects model. The main effects model does not support any claims that sharing face/name behaviors have an effect on the PE of supporting messages. This null result extends to the interaction effects model which does not support any claims regarding RQ3 and joint factual/physical/social disclosures. In my thematic analysis, one participant said, *"On the last one I noticed I believe it was a black lady in the profile pic and was wondering if there was profile pics on the prior ones because I hadn't even noticed or thought about it till then."* Additional models including helper gender and race were fit to contextualize the described phenomenon (omitted due to space constraints). For all dimensions, helpers presenting as non-White were estimated to have a negative effect greater than factually/physically anonymous helpers. My data supports prior work emphasizing that physical identity is inseparable from social identities like gender and race which can lead to negative discrimination [19, 22]. While my experimental design accounted for presented gender and race in physical identity disclosures through randomization, a more focused investigation is required to untangle its effects from physically presented social identities.

3.2 Social Self-Disclosure

RQ2 relates to the first-order effects of revealing a relevant social in-group identity, out-group identity, and withholding those identities. The main effects model supports claims that disclosing recovery status has a small positive effect on the PE of supporting messages. However, the interaction effects model clouds this effect and suggests the effect is local to a subset of participants (see Sec. 3.4). One major qualitative theme bearing a direct relation to RQ2 was **Personalization and Normalization** in disclosing recovery status. Many participants described how socially anonymous comments were generic and lacked personalization, while identifiable in-group comments powerfully related to the original poster's (OP's) situation and normalized their struggle. With respect to the problem-solving dimension of support, these findings mirror prior research emphasizing the utility of support from people with similar health experiences [5, 50]. However, participants described how it can be insensitive for commenters to self-disclose in or out-group identities and draw attention away from the OP. I also found that a subset of participants greatly appreciated the utility of support from a professional. For instance, one said the *"encouragement from the addiction counselor was out of this world."*

3.3 Person-Centeredness

RQ4 pertains to the first-order effects of highly person-centered messages. Here, both main and interaction effects models support claims that explicit acknowledgment of the support seeker's feelings when compared to implicit acknowledgment has a small effect on PE. While still small, HPC had the largest effect with respect to

the problem-solving dimension of support. The qualitative theme **Utility of HPC** also supports this claim. Participants overwhelmingly described how helpful it was to explain the situation, elicit feelings of hope, and *"meet the person where they are."* Similar to Sec. 3.2, participants also described HPC support as personalized, normalizing, and evoking feelings that the helper has *"been there"* in absence of explicit disclosures of recovery status. That said, participants did not unanimously prefer HPC messages, a finding not often reported on in related VPC research. Some participants explicitly objected to support that focused on OPs emotions, especially when those emotions are negative. This is exemplified by one participant who said, *"I think the comment with less is more in this context."* In other words, they preferred the nonfeeling-centered explanation or positive smoothing over present in MPC support. Descriptively, roughly 28% of participants provided higher PE ratings along each supporting dimension for MPC messages than HPC.

3.4 12-Step Norms and Anonymity

My intentions with RQ5 were to investigate the interaction effects between 12-Step norm expectations and contextually relevant identity disclosures. Here, I found significant small negative differences between the ways people in active 12-Step recovery perceive online support from professionals (problem-solving and relational assurance) as well as from socially anonymous support (relational assurance and emotional awareness). Notably, the models produce three significant two and three-way interaction effects regarding 12-Step participation and factual/physical identity. I am cautious of making direct claims about these effects given the potential confounds discussed in Sec. 3.1. One explanation relates to norms surrounding gender and 12-Step support often characterized by same-gender 12-Step sponsorship [73]; one participant epitomized these norms and said, *"I don't think it's ever a good thing for a male to reach out to a female, or vice versa, in recovery, for advice."* The theme **12-Step Polarization** shows that 12-Step ideas like sponsorship and one's program can be isolating. While a number of participants wanted a stronger emphasis on step work and attending meetings, others explicitly acknowledge their negative biases related to 12-Step. One said, *"I guess I am just sour on the whole AA lingo and culture."* Prior work has investigated the wide variety of client and clinician perspectives towards 12-Step programs [45, 68] and compared the efficacy of 12-Step programs to their alternatives [74]. Importantly, my findings show that these different perspectives impact people's perceptions of identity in online support.

3.5 Cross Cutting Themes

While not related specifically to any particular RQ, here I present two cross-cutting themes to provide additional insight. First, the most salient theme in my analysis was **Recovery is Hard**. Participants often described how they could relate to the pain of the support seeker and mentioned that any support is great. Specifically, support that emphasized the seeker is not alone and communal aspects of recovery non-specific to group affiliation. When comparing HPC and MPC messages, many mentioned how similar they were in regards to positivity, empathy, and sincerity. One participant noted, *"Some comments do not matter more than others, they all try to accomplish the same thing."* Second, and most surprising to me,

Table 1: A table describing the model coefficients and related estimated error for 3 orbetareg models relating to distinct support dimensions. Bold values denote statistical significance when 95% confidence intervals do not include 0. All non-intercept parameters are categorical factors where italicized parameters relate to participants and non-italicized parameters relate to experimental conditions. The main effects model is omitted due to space constraints.

Parameter	Estimate (Est. Error)		
	Helpful	Supportive	Sensitive
<i>Subject Intercept</i>	0.02 (0)	0.02 (0)	0.02 (0)
<i>Intercept</i>	0.15 (0)	0.15 (0)	0.15 (0)
<i>Woman</i>	0.06 (0.03)	0.07 (0.02)	0.06 (0.03)
<i>12-Step</i>	0.03 (0.04)	0.04 (0.03)	0.01 (0.04)
HPC	0.05 (0.01)	0.03 (0.01)	0.02 (0.01)
Factual Iden.	0.02 (0.02)	0.01 (0.02)	0.01 (0.02)
Social Anon.	-0.02 (0.02)	0.01 (0.02)	-0.02 (0.02)
Social Out	0.02 (0.02)	0.01 (0.02)	-0.01 (0.02)
12-Step:Factual Iden.	-0.07 (0.04)	-0.09 (0.05)	-0.06 (0.04)
12-Step:Social Anon.	-0.03 (0.04)	-0.1 (0.05)	-0.1 (0.05)
12-Step:Social Out	-0.08 (0.04)	-0.12 (0.05)	-0.05 (0.04)
Factual Iden:Social Anon.	-0.02 (0.03)	-0.01 (0.03)	-0.01 (0.03)
Factual Iden:Social Out	-0.04 (0.03)	-0.02 (0.03)	-0.02 (0.03)
12-Step:Factual Iden:Social Anon.	0.06 (0.05)	0.08 (0.04)	0.09 (0.04)
12-Step:Factual Iden:Social Out	0.05 (0.05)	0.06 (0.04)	0.02 (0.05)

was participant desires for **Actionable Support**. My experimental investigation was directed at emotionally supportive interaction in OHC and as such support seeking messages were written to elicit emotional support. However, participants often noted a comment's lack of actionable support in the form of coping strategies and situational advice. One participant indicated an explicit tension between this desire and 12-Step norms and said, "*neither offered much in the ways of help, although I know it is frowned upon to give advice during shares or cross talk*"². This theme echoes prior work emphasizing the importance of support match [56, 69].

4 DISCUSSION

Based on my findings, I identified three implications to inform future research regarding anonymity, online support, and to motivate the moderation and design of OHC.

4.1 Mixed OHCs - Implications for Design and Research

Future work should consider organizational differences stemming from social group sub-identities within all-inclusive health contexts. My findings highlight this contrast in the context of online recovery. This phenomenon of institutions driving online values has recently been documented in the context of online gaming and similarly poses a challenge to game designers [63, 75]. OHCs that don't acknowledge these differences may encounter issues. Similarly, online recovery research has often focused on 12-Step groups [30, 58, 73]. I acknowledge the influence of this work in shaping the field's understanding of online recovery while proposing future research in this space should include participants outside of 12-Step

recovery to develop inclusive ways to support universal desires for community in recovery.

4.2 Emotionally Supportive Interaction - Implications for Research

My investigation points to a number of findings related to the ecology of sensitive online support. Some support research, mine included, directs their investigations solely on emotional support [40, 42]. My findings suggest that expectations of informational support often accompany emotional support seeking. To better relate to real OHCs, I recommend future work include emotional and action-focused support consistent with other investigations [36, 46]. To my knowledge, this is the first investigation of how VPC messages are perceived by people outside the immediate supportive interaction. My findings support the use of VPC to evaluate secondary participant benefits of supportive interactions and show similar positive perceptions. This is relevant to exploring the network effects of VPC support online and further support the positive effects of lurking behaviors in OHCs [67]. However, researchers should be cautious of varying opinions of correctness given that VPC only relates to the acknowledgment of the support seeker.

4.3 Lessen Support Provisioning Barriers - Implications for Design

Finally, work should continue to find new ways to support helpers in OHC and help them navigate tensions, including those surrounding anonymity, that may pose barriers to support provisioning. While still being sensitive to other desires/expectations for support, participants overwhelmingly approved of all forms of support in the present study and the apparent effects of each condition are notably small. Considering prior research relating time spent in an OHC to interpretations of anonymity [58], my findings highlight

²In 12-Step, cross talk is when someone engages directly with stories shared by other members [1]

the importance of being flexible to the desires and concerns of new support providers as they learn and adapt to specific OHC values.

4.4 Limitations

The presented work allowed me to explore the effects of anonymity, VPC, and 12-Step participation on PE of supportive interaction in recovery but also has inherent limitations. I must acknowledge the risks associated with using self-reported data to identify 12-Step participants. Additionally, the efficacy of sliding scales has received mixed results [7, 26, 35, 70, 71] and while I attempted to mitigate these concerns through usability pilots and careful monitoring of study communication channels, it is possible the decision biased my results. Finally, while a number of participants mentioned the realism of messages in the study, it is possible the lab setting of this investigation does not directly generalize to real OHCs and non-recovery oriented communities. Real OHCs are individually unique [34] and future research should extend this investigation by analyzing the effects of VPC support and identity as it naturally occurs online.

5 CONCLUSION

People in online recovery communities present a unique perspective on the diverse effects of online anonymity in support. I conducted a mixed-method analysis to understand how enacted online anonymity affects the perceived effectiveness of supportive interaction. From my findings, I conclude the effects of enacted facets of anonymity in online recovery are small and mostly explained by organizational differences between 12-Step groups and their alternatives. Considering this, I provide implications for research on VPC and emotional support and, propose design implications for sensitive online contexts and support provisioning.

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